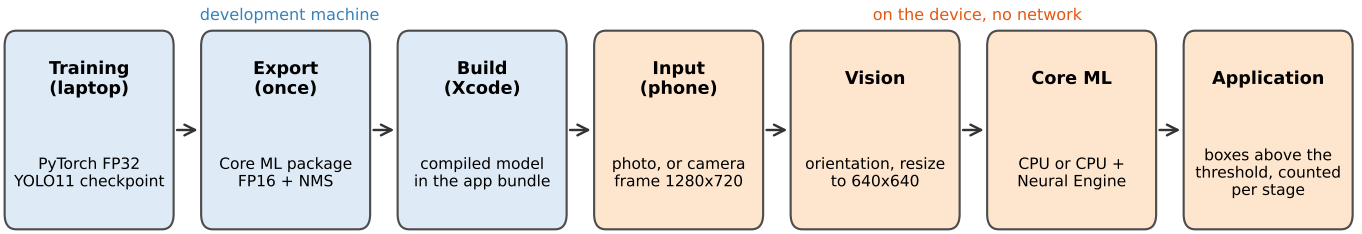
development machine
on the device, no network
Training (laptop)
PyTorch FP32
YOLO11 checkpoint
Export (once)
Core ML package
FP16 + NMS
Build (Xcode)
compiled model
in the app bundle
Input (phone)
photo, or camera
frame 1280x720
Vision
orientation, resize
to 640x640
Core ML
CPU or CPU +
Neural Engine
Application
boxes above the
threshold, counted
per stage